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Purpose/Policy:

The purpose of the “Warmer” standard operating procedure is to help assist with the setup, operation, changeover, troubleshooting, and CIP.

Scope:

The Warmer is a long tunnel that uses steam to warm up cans to help pressurize them and help mitigate condensation. Cans are moved through the warmer by a long conveyor belt and warmed up with steam that comes from several different locations throughout the warmer. Before start up make sure the E-Stop is pressed and the valve for steam located behind the warmer is in the off position.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- E-Stop – Emergency Stop button
- CIP – Cleaning in Process
- F400 – Caustic cleaning solution (Rochester Midland)
- UHMW – Ultra-High Molecular Weight plastic (used for guides and wear strips)
- Warmer – Steam tunnel that warms filled cans to help mitigate condensation

Authorities/Responsibilities:

This SOP can be completed by an entry level packaging operator after proper training from a senior member of the Can line packaging team. The objective of this SOP is to gain the knowledge needed to operate the “Warmer” and trouble shoot any issues that may arise.

Resources and Materials:

Operation:

- No equipment required

Changeovers:

- UHMW spacer blocks
- Empty cans

Basic Cleaning:

- Face Shield
- Green Rubber Chemical Gloves
- Safety Glasses
- F400 Foamer

Deep Clean:

- Refer to SOP Document # CL###
    - Link to document goes here.

Troubleshooting:

- No Equipment required.

PPE Required:

- Face Shield
- Green Rubber Chemical Gloves
- Safety Glasses

Procedures:

Setting up the Warmer:

1. To start up the warmer first walk behind it and turn on the valve located on the insulated water pipe. This will turn on steam to the warmer.
2. Make sure the filter screens are in the holding tank.
3. Close the valve at the bottom of the warmer to close the drain.
4. Pull out the E-Stop
5. Press the “Control Power” button.
6. Next press the green start button to start the warmer.

Operation of the Warmer:

1. Once the warmer is powered on and running the next step is to make sure the conveyors are on.
    1.1 Refer to Conveyor SOP
2. Make sure the warmer is set to the correct temperature to prevent condensation.
    2.1 Refer to the “Warmer Relativity Chart”.
    2.2 If you need to adjust the temperature, use the up and down arrow buttons below readout.

Changeover Guide:

1. The first step to change over the warmer is to loosen the hand knobs at the infeed of the warmer.
2. Lift up or lower the plate so that an empty can of the size you are changing over to can fit underneath the plate without tipping over.
3. Next tighten down the knobs that were previously loosened.

4. To adjust the outfeed of the warmer, loosen the hand knobs that are on both sides of the warmer and on top of the UHMW.
5. Lift the outfeed blower and either place blocks above or below the brackets depending on which size you are changing over to.
6. Tighten down the hand knobs on the outfeed.

Shutting Down Warmer:

1. Make sure all cans have made it through the warmer before starting shut down.
2. Walk behind the warmer and turn off the steam valve located on the insulated pipe.
3. Open the valve at the bottom of the water holding tank and drain all the water.
4. Press stop on the control panel.
5. Press E-Stop.

Basic Cleaning of the Warmer:

1. Nightly Cleaning
    - 1.1 Make sure the water is drained out of the warmer.
    - 1.2 At the end of every workday, pull out screens from the water holding tank.
    - 1.3 Spray the screens and the inside of the water holding tank with F400 foam.
    - 1.4 Let the foam sit for 10 minutes and then rinse off with hot water.
    - 1.5 Once rinsed off with water place screens back inside warmer and let air dry.
2. End of Week CIP instructions
    - 2.1 To start the Warmer up for CIP following the following steps.
        - 2.1.1 Close the water holding tank valve underneath the warmer.
        - 2.1.2 Turn on the steam valve located on the back of the warmer (insulated pipe).
        - 2.1.3 Verify, the water valve located behind the warmer is in the open position. If closed, then turn it to the open position.
        - 2.1.4 Turn on the main power located on the control panel.
        - 2.1.5 Reset the red E-Stop button.
        - 2.1.6 Press the control power button.
        - 2.1.7 Press the green start button.
        - 2.1.8 Manually turn up the temperature of the water to 135 degrees using the up-arrow keys on the block temperature control box located on the main control panel.
        - 2.1.9 On the control panel use the panel view C1000 conveyor controls to turn on the console by pressing the green reset button.
        - 2.1.10 Go to the manual screen and turn on the conveyor M12, M13, and M14.
    - 2.2 Now that the warmer is running, we can start the CIP

2.2.1 Wearing the appropriate PPE (Safety shield, Safety glasses, green rubber gloves) retrieve ½ gallon of caustic.
2.2.2 Verify that the warmer has reached 135 degrees F.
2.2.3 Carefully pour the caustic into the water holding tank located on the front of the warmer (Wear face shield and green rubber gloves).
2.2.4 Turn on the spray pump switch from the control panel located on the warmer by turning the switch to H (Hand/manual).
2.2.5 Let the caustic cycle for 20 minutes.

2.3 After the 20-minute cycle we will drain the warmer.
2.3.1 Press the red stop button on the control panel.
2.3.2 Open the main drain underneath the water holding tank on the warmer.
2.3.3 Verify the water has been drained.
2.3.4 Close the drain valve on the water holding tank.

2.4 After draining the warmer we will want to rinse it.
2.4.1 Fill up the warmer again by pressing the green start button on the control panel.
2.4.2 Let it run for 10 minutes.
2.4.3 Drain the warmer by following steps 2.3.1 through 2.3.4.
2.4.4 On the control panel manually turn down the temperature to 75 degrees using the down arrow key on the temperature control block.
2.4.5 Follow steps 2.4.1 through 2.4.3 again.
2.4.6 Follow steps 2.3.1 through 2.3.3 for a second time to drain the warmer again.
2.4.7 Rinse the water holding tank with a hot water hose.
2.4.8 This will ensure all the chemical has been diluted and rinsed out to an acceptable level.

2.5 We will then run an acid cycle after rinsing out the caustic.
2.5.1 Close the water brink drain valve.
2.5.2 Fill up the warmer again by pressing the green start button on the control panel.
2.5.3 Wearing the appropriate PPE, retrieve ½ gallon of acid.
2.5.4 Verify the warmer has reached 75 degrees F.
2.5.5 Carefully pour the acid into the water holding tank on the front of the warmer. (Wear face shield and green rubber gloves)
2.5.6 Verify that the spray pump is on using the switch on the control panel by turning the switch to H (hand/manual).
2.5.7 Cyle the acid for 20 minutes.
2.5.8 Drain the warmer by following steps 2.3.1 through 2.3.4

2.5.9 On the control panel manually turn the temperature back up to 95 degrees F. using the up-arrow key on the temperature control block.

2.5.9.1 Note: 95 F. is the original operational set point.

2.6 Once the acid cycle is complete, we will shut down the warmer.

2.6.1 Turn off the warmer by pressing the red stop button on the control panel.

2.6.2 Press the E-Stop button.

2.6.3 Turn the main power off on the warmer control panel.

2.6.4 Shut the steam valve located behind the warmer (insulated pipe).

2.6.5 Rinse off running conveyors.

2.6.6 Press the red stop button on the Panel View C1000 to shut off the conveyors.

2.6.7 The warmer has now finished its CIP.

Deep Cleaning:

Outline the deep cleaning process for the SOP equipment. See below for example:

1. Refer to SOP Document # CL###
   1.1 Link to document goes here.

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

1. If the warmer isn't getting up to temperature check with someone from maintenance to make sure the boiler is on.
2. If the water storage tank is overflowing, make sure high water level floats are still in place.

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

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Approved by: Kyle Szymanski

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